JOHN PATRICK COLLINS, M.S.

BIOINFORMATICS DATA SCIENTIST | SCIENTIFIC SOFTWARE ENGINEER | AI-NATIVE COMPUTATIONAL BIOLOGY

(REMOTE U.S.)

PROFILE

- **Bioinformatics scientist and scientific software engineer** translating novel assays and complex biological data into validated, reproducible workflows and scientist-facing products across clinical diagnostics, immuno-oncology, single-cell genomics, and long-read synthetic biology.
- **Advanced Python developer** with end-to-end ownership from requirements and data modeling through implementation, benchmarking, QC, documentation, and operational handoff.
- Hands-on with **NGS/ONT, Benchling/LIMS, cloud/HPC,** and **supervised AI-assisted engineering.**

EXPERIENCE

INDEPENDENT COMPUTATIONAL BIOLOGY
Researcher / Open-Source Developer

2025 - PRESENT

Lead CGT-to-CORD research testing whether complex perturbation responses can be represented by compact, biologically interpretable axes across 40+ public single-cell datasets, with orthogonal CRISPR gene-fitness and TCGA tumor-state evaluations. Architect reusable Python/Colab workflows with versioned inputs, content-hash caches, deterministic scoring, run/release manifests, claim-evidence ledgers, and leakage-aware held-out evaluation. Use frontier coding/reasoning agents for supervised implementation, testing, debugging, review, and documentation.

BACSTITCH DNA, INC.
Staff Software Developer

MARCH 2024 - NOVEMBER 2024

Co-designed and implemented Nanoplex, the Python/ONT pipeline used to verify SCRIVENER assemblies: combinatorial-barcode demultiplexing, minimap2/samtools alignment, bcftools/Sniffles2 variant calling, SciPy read-length clustering, and automated QC/purity scoring on a 36-core cluster.[1] Analyzed sequence-verification data for a Cell Systems study of >5,000 DNA assemblies (2-19 blocks; constructs to 81 kb), generated publication evidence, edited the manuscript, and coauthored the paper. Built Benchling API tools and data models for assembly planning, sample lineage, inventory/entity creation, and LIMS registration; partnered across R&D and software on infrastructure and CI planning.

CALICO LIFE SCIENCES
Data Management Engineer, Contract

SEPTEMBER 2023 - NOVEMBER 2023

Built Python ETL, normalization, and validation automation for migrating laboratory inventory into Benchling, preserving consistent entity representations and enabling reliable downstream lab-management workflows.

CONTACT

LOCATION: Colorado Springs, CO
PHONE: **(820) 465-9453**
EMAIL: jcollins.bioinformatics@gmail.com
LINKEDIN: linkedin.com/in/johncollins-bioinformatics
WEB: johnpatrickcollins.info
GITHUB: github.com/jcollins-bioinfo

EDUCATION

2013 - 2014 | **BIOMOLECULAR ENGINEERING AND BIOINFORMATICS (M.S.),**
University of California, Santa Cruz

2009 - 2013 | **BIOENGINEERING (B.S.),**
Graduated With Honors in the Major,
University of California, Santa Cruz

CORE EXPERTISE

SCIENTIFIC SOFTWARE. Python (NumPy, SciPy, pandas, Polars, scikit-learn), R, Bash/Linux, Git; pipelines, APIs, ETL/data models, and testing.

COMPUTATIONAL GENOMICS. NGS/ONT, RNA-seq/scRNA-seq, cfDNA, V(D)J; demultiplexing, alignment, assembly, variant calling, and QC.

VALIDATION & ANALYSIS. Statistical modeling, clustering, permutation tests, benchmarking, leakage-aware evaluation, and reproducibility.

PLATFORMS & APPLICATIONS. Benchling LIMS/ELN/API; AWS, Azure, GCP; Docker; Linux/HPC; Dash/Plotly and Flask.

AI-ASSISTED ENGINEERING. Specifications, task decomposition, code review, tests/evals, provenance, and human verification.

PEER-REVIEWED PUBLICATIONS

[1] Matsui T, Hung P-H, Mei H, Liu X, Li F, Collins J, et al. **High-throughput DNA engineering by mating bacteria.** Cell Systems. 2026:101656. doi:10.1016/j.cels.2026.101656

[2] Grskovic M, Hiller DJ, Eubank LA, Sninsky JJ, Christopherson C, Collins JP, et al. **Validation of a Clinical-Grade Assay to Measure Donor-Derived Cell-Free DNA in Solid Organ Transplant Recipients.** J Mol Diagn. 2016;18(6):890-902. doi:10.1016/j.jmoldx.2016.07.003

Experience (Continued)

PACT PHARMA, INC.
Bioinformatics Data Scientist
SEPTEMBER 2018 - SEPTEMBER 2020
Developed Python-centric NGS and Sanger analysis software for personalized adoptive T-cell immunotherapy and a patient-specific phase I clinical program.[3] Designed Benchling chain-of-identity and automated data-capture workflows linking patient, sample, construct, sequencing, and analysis records. Built a Dash application for gene-edited plasmid sequencing that reduced routine processing from hours to minutes and delivered interactive QC and results to scientists.

BRISTOL MYERS SQUIBB
Bioinformatics Pipeline Developer, Contract
FEBRUARY 2018 - JUNE 2018
Integrated open-source bioinformatics tools with custom Python into a parallelized V(D)J gene-rearrangement analysis application for an HPC environment.

BIO-RAD LABORATORIES
Bioinformatician II - Sequencing Analysis
APRIL 2017 - JULY 2017
Developed secondary and tertiary NGS analysis for a single-cell whole-transcriptome RNA-seq assay and automated cloud-based analytical reports using the Python scientific stack.

CAREDX, INC.
Bioinformatics Analyst
NOVEMBER 2014 - APRIL 2017
Engineered Python R&D software and operated NGS pipelines for donor-derived cell-free DNA, supporting analytical validation and assay-performance development for AlloSure, a noninvasive solid-organ transplant diagnostic.[2,4,5] Performed MiSeq library preparation for PCR-based dd-cfDNA quantification, connected wet-lab observations to pipeline QC, statistical analysis, and interpretation, and coauthored the clinical-grade assay validation paper.

UNIVERSITY OF CALIFORNIA, SANTA CRUZ
Undergraduate & Graduate Student Researcher
2011 - 2014
Contributed to single-cell RNA-seq, nanopore sequencing, and chromosome-protein dynamics research in the Pourmand, Akeson, and Bhalla laboratories, respectively.

Methods & Tools

GENOMICS. minimap2, BWA, samtools, bcftools, Sniffles2, GATK, Dorado, Clustal Omega; UCSC Genome Browser, Ensembl, and NCBI.

DATA & ML. NumPy, SciPy, pandas, Polars, scikit-learn; statistical modeling, clustering, permutation tests, representation learning, experimental design, and scientific visualization.

DELIVERY. Requirements, implementation, benchmarking, automated QC, documentation, cross-functional review, and operational handoff.

AI & Reproducibility

Direct frontier coding and reasoning agents through explicit specifications, task decomposition, implementation, debugging, review, tests/evals, and documentation while retaining human ownership of architecture, biological interpretation, falsification, security boundaries, provenance, and scientific claims. Reproducible research practice includes versioned inputs, content-hash caches, deterministic scoring, run/release manifests, claim-evidence ledgers, and held-out evaluation.

Published Conference Abstracts

[3] Dalmas O, Pan Z, Shieh C, et al., including Collins J. **A high-throughput platform to produce neoE-HLA libraries for capturing neoE-specific T cells from the peripheral blood of patients with solid tumors.** Cancer Res. 2020;80(16 Suppl):Abstract 3253. doi:10.1158/1538-7445.AM2020-3253

[4] Crespo-Leiro M, Hiller D, Woodward R, Grskovic M, Marchis C, Song M, Collins J, Zuckermann A. **Analysis of Donor-Derived Cell-Free DNA with 3-Year Outcomes in Heart Transplant Recipients.** J Heart Lung Transplant. 2017;36(4):S69-S70. doi:10.1016/j.healun.2017.01.172

[5] Khush KK, Grskovic M, Luikart H, et al., including Collins J. **Circulating Cell-Free DNA as a Non-Invasive Marker of Pediatric Heart Transplant Rejection and Immunosuppressive Treatment.** J Heart Lung Transplant. 2016;35(4):S75. doi:10.1016/j.healun.2016.01.205